

YUEQI XU

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EDUCATION

University of Washington Seattle, WA
Ph.D. in Biosatistics *Sep. 2023 – Present*

University of Washington Seattle, WA
Master of Science in Biosatistics *Sep. 2022 – Jun. 2023*

- Transferred to Ph.D. program

University of Washington Seattle, WA
Bachelor of Science in Statistics *Sep. 2017 – Mar. 2022*
Minor in Informatics & Mathematics
Honors: Dean's List, Graduated Magna Cum Laude

PROJECTS & RESEARCH

Analytical Methods for Compositional Data with Applications in the NHANES Study *Jun. 2024 – Present*
Supervisor: Dr. Chongzhi Di

- Developed statistical tools for compositional data analysis to flexibly model dose-response relationships between physical activity patterns and health outcomes.
- Conducted simulation studies to evaluate and compare the performance of the proposed methods against established techniques, such as isometric log-ratio-transformed regression and isotemporal substitution models.
- Applied the developed methods to physical activity data from the National Health and Nutrition Examination Survey (NHANES) study.

Development and Comparison of Multiple-Testing Methods *Jul. 2023 – Present*
Supervisor: Prof. Gary Chan

- Reviewed and analyzed existing methodologies for multiple testing, with a focus on large-scale partial conjunction tests.
- Developing hypothesis-testing procedures for large-scale partial conjunction test aimed at improving testing power while controlling the false discovery rate (FDR) at desired levels.
- Designed and implemented simulation studies to evaluate and compare the performance of existing methods against the proposed approaches in various high-dimensional settings.
- Investigated applications to mediation analysis and reproducible signal detection across multiple studies or conditions.

CLASS PROJECTS

Predicting Student Performance in Game-Based Education *Apr. 2023 – Jun. 2023*
Instructor: Professor Brian Leroux

- Utilized logistic regression, random forest, and gradient boosting algorithms to predict student performance in an educational game using game logs.
- Conducting a thorough evaluation and comparison of each method's performance.

A Comparison of Clustering Methods *May. 2023 – Jun. 2023*
Instructor: Professor Marina Meila

- Implemented **mean shift clustering** and **hierarchical clustering** algorithms on a given dataset using **R**, comparing their performance through visual assessment, stability analysis, and internal evaluation.

Review on A Spatial Analysis of Multivariate Output from Regional Climate Models *Feb. 2022 – Mar. 2022*
Instructor: Professor Vincent Roulet

- Conducted a comprehensive literature review on hierarchical spatial spatial analysis method of multivariate output from Regional Climate Models (RCMs).

A Predictive model for Perfume Prices

Jan. 2022 – Mar. 2022

- Conducted **exploratory data analysis** (EDA) and developed **predictive models** in **R** to analyze key factors contributing to retail perfume prices. Employed **step-wise backward selection** to identify the best model and evaluated model performance using **cross-validation**.

Method Review of the BNT162b2 Vaccine Phase III Trial

Instructor: Professor Ranjini Grove

May. 2021 – Jul. 2021

- Analyzed the **Bayesian work-flow**, specifically the prior belief and parameter transformations, proposed in the efficacy trial of BNT162b2, the Pfizer-BioNTech Covid-19 vaccine. Visualized analytical results with **ggplot2**.
- Developed alternative justifiable Bayesian work-flows, emphasizing our favored philosophy of prior selection, and utilized **grid approximation** to determine parameters that satisfy the proposed prior belief.

TEACHING EXPERIENCES

University of Washington, Department of Biostatistics

Teaching Assistant

Sep. 2023 - Dec. 2024

- BIOST 511: Medical Biometry I
 - Instructor: Prof. Lloyd Mancel*Autumn 2023*
- BIOST 512: Medical Biometry II
 - Instructor: Prof. Lurdes Inoue*Winter 2024*
- BIOST 513: Medical Biometry III
 - Instructor: Prof. Susanne May*Spring 2024*
- BIOST 310: Medical Biometry III
 - Instructor: Prof. William Brown*Autumn 2024*

SKILLS

Statistical Analysis: Regression, Longitudinal Data Analysis, Survival Analysis, Functional Data Analysis, Compositional Data Analysis, Bayesian Analysis, Non-parametric Analysis, semi-parametric analysis, Time Series, Inferential Statistics

Programming Languages

- Proficient: R
- Advanced: SQL, Python
- Basic: Java, JavaScript, HTML5, CSS

Software and Tools

- Proficient: LaTeX, R Markdown
- Advanced: Excel, Word, PowerPoint, Git
- Basic: Tableau, d3.js

Soft Skills: Communication, Collaboration, Time Management, Planning, Critical Thinking, Problem-Solving

PROFESSIONAL ASSOCIATIONS

American Statistical Association

2019 – Present